

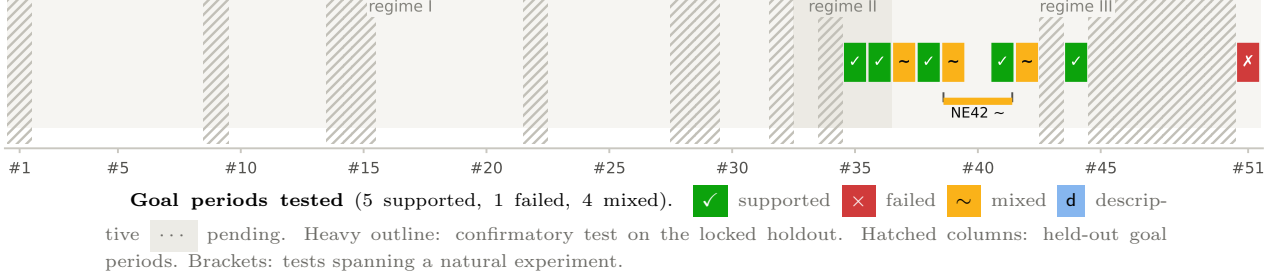
H47: Rooms set the coherence length

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2.5/5

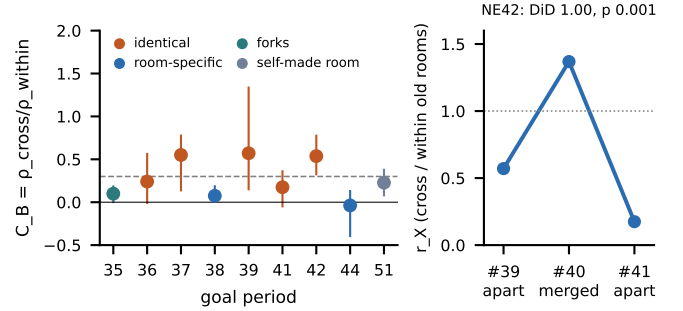
Bottom line. Content coherence in the village stops at room boundaries when rooms work on different things and moves with the chat channel at merges and splits, but inside rooms it follows conversation, and no room leads a goal change.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

HH172 claimed rooms set the swarm's coherence length: content co-moves inside a room but not across, so a per-room topic detector beats the swarm one, and one room leads at goal changes. We measured per-pair content correlation within (ρ_w) and across (ρ_c) rooms in 30-min windows on 9 multi-room periods, against room-relabel nulls. Then three native tests: NE42's merge/split, #41's identical instructions, and #51's #focus room. Synthetic worlds came first. Rooms bound coherence: $C_B = \rho_c / \rho_w \leq 0.3$ in 6/9 periods (median 0.23). Merging #best and #rest merged their content fluctuations; splitting restored the boundary (DiD 1.0, p 0.001). But inside rooms, correlation sits in pairs that talk (and the post hoc boundary sharpness tracks topic separation). No per-room detector gain (2 clean events). No leader: both rooms switch within minutes. **Mixed.** Holdout script written, not run.



SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Soft-spin $O(n)$ vector field (model 11) on a room graph. Spin s_i : an agent's 32-d whitened statement mean per 30-min window, minus its unit mean and static goal directions. Distance is categorical: same room, different room, and inside a room high vs low mention tiers. Drives: global (goal, schedule), room-specific (instructions, tasks); coupling through the room's chat. S_i : split-half signal variance.

$$\rho_{w,c} = \frac{\sum_{\text{pairs}} \langle \delta s_i \cdot \delta s_j \rangle}{\sum_{\text{pairs}} \sqrt{S_i S_j}}, \quad C_B = \frac{\rho_c}{\rho_w}, \quad G = \frac{\rho_{w,\text{low}}}{\rho_{w,\text{high}}}$$

$$r_X = \frac{\rho(\text{cross old rooms})}{\rho(\text{within old rooms})}, \quad L = \langle y_{\text{best}}(\tau) - y_{\text{rest}}(\tau) \rangle_{\tau < 2h}$$

A useful, if unsurprising, design rule: rooms are content firewalls only when they do different things. Same task in two rooms shares about half of each room's content fluctuation; different tasks share almost none. Merging rooms merges coherence within days, and splitting restores it at once, so the chat channel matters (NE42). Inside a big room, coherence lives in conversation pairs, so a 25-agent room is not one unit to monitor. The two dynamic claims failed cleanly. Goal changes hit every room within minutes, so there is no leader to watch. A per-room topic alarm adds nothing to H36's swarm alarm at the events we have. Caveat for steering: C_B cannot say whether rooms couple or are just driven by their tasks.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 $C_B \leq 0.3$, $p < 0.05$ in $\geq 2/3$	6/9 (3 mixed); median 0.23	passed
P2 flat inside rooms ($G \geq 0.6$)	median G 0.40; #51 one room failed (half) 0.18	failed (half)
P3 per-room R1 beats swarm R1	AUC 0.62 vs 0.79; clean events: none fire	failed
P4 no room leads	1/8 sig. (#38, a level offset); Stouffer p 0.38	passed
P5 #41 sharp with same instructions	C_B 0.17, p 0.003	passed
P6 instruction direction carries $\leq 15\%$	-0.02, -0.01 (#38, #44)	passed
P7 NE42 boundary follows channel	DiD 1.00 [0.50, 1.34], p 0.001; #39 above 0.3 met)	mixed (DiD met)
P8 #focus cuts its members	already decoupled before (0.08 / 0.27 / 1.33)	failed
Post hoc: C_B vs separation	Spearman -0.83 (p 0.014, 8 peers)	lead

GOAL PERIODS IN DETAIL

G35	✓ supported
G36	✓ supported
G37	~ mixed
G38	✓ supported
G39	~ mixed
G41	✓ supported
G42	~ mixed
G44	✓ supported
G51	✗ failed
NE42	~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

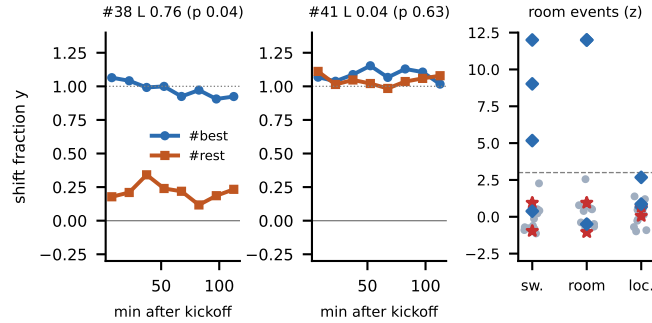
CAVEATS

C_B is the global share of a room's fluctuation. Coupling and room-specific task drives both lower it, and the synthetic shows they cannot be separated. Mention tiers are endogenous (co-workers both talk and share topics). NE42 is one goal-confounded A-B-A with 15 agents. The detector test has 2 clean events and 11 placebo days. The quench completes within 1–3 min, below the 15-min bins. Units are 1–17 days long, P1 passes exactly at its threshold, and the post hoc patterns are leads. One embedding model; style residuals checked.

ROUND 2

Round 2: test coupling directly with DQ1's context ledger (does content move toward room messages an agent actually read?), and run the confirm script (#45–#47, #50, NE15) after the reuse disclosures.

- **Essence.** where the boundaries of shared content are in an agent swarm, and whether they are set by the channel (who can read whom), the task, or conversation. Round 1: the channel bounds it (NE42), the task sets how sharp the boundary is, and conversation shapes it inside a room.
- **H47-R1** Read-out-conditioned coupling (the mechanism test).
- **H47-R2** Coherence length inside rooms on the reply graph.
- **H47-R3** Confirmatory run (confirm.py: #45–#47, #50, NE15) after the reuse disclosures, and coordination with H26's confirm (shared targets).
- **H47-R4** Instrument: DQ5's gte embedding swap and the final style-residualized vectors.
- **H47-R5** Drop or redesign leadership.
- **H47-R6** Detector: per-room R1 can only be evaluated on more room events; the onboarding and showcase rooms inside the holdout windows (#46, #50) are the remaining ones.



Second figure. Left, middle: kickoff response (shift fraction y , 15-min bins). Both rooms are switched from the first bin. #38's significant lead index is a level offset, not a lag. Right: room-event scores (stars clean, diamonds goal-confounded) against multi-room placebo days (gray) for the swarm, per-room and localized detectors. Only the goal-confounded events fire.